
Position: Verifiable Data Minimization is a Prerequisite for Responsible, Privacy-Preserving Industrial Vision

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Abstract

The adoption of computer vision to drive industrial efficiency and safety creates a persistent tension between operational utility and worker privacy. Current privacy measures, such as post-hoc blurring, are fundamentally flawed: they depend on the error-prone detection of sensitive attributes yet do not prevent the data from being repurposed for unintended inferences. We posit that industrial computer vision must shift from “hiding secrets” to verifiable data minimization. Formalized through the Information Bottleneck principle, we advocate for sensing pipelines that are optimized to capture only the features necessary for a declared task (e.g., pose estimation), without requiring labels for what is sensitive. This bounds the system’s capacity for unauthorized inferences such as re-identification or function creep. We outline an implementation path using modular edge processing and trusted execution environments, and identify key research challenges for making verifiable purpose limitation a prerequisite for responsible industrial AI.

1. Introduction

The adoption of AI-powered computer vision in industrial settings is central to Industry 4.0 (Jagatheesaperumal et al., 2022) and a key enabler of the human-centric vision of Industry 5.0 (Martini et al., 2024). While these systems promise gains in efficiency, safety, and worker welfare, their deployment increases the risk of automated extraction of privacy-sensitive information from visual data (Schröder et al., 2025). This creates a persistent tension between video-driven process monitoring and worker privacy (Ogrezeanu et al., 2022; Mannhardt et al., 2019). Stakeholders face trust

and compliance barriers, workers are uneasy about monitoring (U.S. Government Accountability Office, 2024), and companies must protect privacy and sensitive site data (Paradise Vit et al., 2024). The richness of visual data further enables function creep, whereby footage collected for one purpose may be repurposed for others without explicit consent (Koops, 2021). Through the lens of contextual integrity (Nissenbaum, 2004), these risks arise from inappropriate information flows: while consent is only given for a specific purpose, the data captured can be used for many others.

Consider a manufacturer that installs cameras on an assembly line to monitor worker posture for ergonomic assessment. The declared task is pose estimation, but captured frames inevitably contain far more: worker faces and clothing that enable re-identification, hand-held tools that reveal production processes, and the facility layout, which may constitute intellectual property. Whether footage is sent to a cloud API or processed on-premises, any system with access to raw frames retains the capacity for inferences the worker never consented to.

While computer vision privacy research has advanced dataset privacy and private training of models, it has paid less attention to inference-time safeguards once models are deployed (Miresghallah et al., 2020). In reality, inference privacy is often handled by using redactive obfuscation techniques tackling sensitive regions (e.g., blurring faces or bodies) (Zhao et al., 2025). We argue these systems are fundamentally misaligned with the core privacy risks of function creep and unintended information leakage. They attempt to secure data by hiding or stripping sensitive elements before downstream processing, but they do not inherently limit what information is retained to the strictly necessary. This leaves systems vulnerable to repurposing and unauthorized extraction of sensitive information, as they can still encode non-consented information that may be extracted later.

What is fundamentally missing is not better concealment of sensitive data, but an architectural mechanism to bound and quantify information retention. This aligns with Privacy by Design (PbD) (Cavoukian et al., 2009), which treats privacy as a proactive default rather than a post-hoc patch. At its core is Data Minimization, mandated by GDPR (European

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Parliament & Council of the European Union, 2016) (Art. 5(1)(c)) and CPRA/CCPA (Legislature, 2018): collect only what is *adequate, relevant, and necessary* for a declared purpose (European Data Protection Board, 2020). Crucially, minimization is task-relative: excess depends on the intended inference. For industrial vision, a system designed for posture analysis should constrain its representation to encode only task-relevant information. Yet, technical mechanisms to enforce this legal mandate at the architectural level are largely absent (Ganesh et al., 2025).

We take the position that industrial computer vision must include verifiable purpose limitation through strict data minimization. Systems should be restricted, via mathematical optimization and secure hardware, to retain only task-sufficient information, thereby provably limiting their capacity to encode non-consented information beyond what is implied by the declared task.

Designing systems to capture only task-sufficient information ensures non-consented data never leaves the camera. This shifts the paradigm: the sensing pipeline acts as a privacy filter, eliminating spurious identifying visual characteristics (facial detail, clothing, bystanders, spatial context) that are orthogonal to the task, while retaining only essential task signals (e.g., skeletal pose) before they leave the camera. Because sensitive attributes can be inherently entangled with these task signals (e.g., gait and identity), the extent of information removal ultimately depends on this correlation. Our approach therefore shifts the focus from binary privacy guarantees to bounding and auditing residual risk.

In this paper, we formalize strict data minimization for task-bound vision using information-theoretic principles, and examine what implementing it in practice actually requires. We identify key challenges and requirements, and outline an implementation blueprint using modular edge processing and Trusted Execution Environments (TEEs). We situate the approach relative to existing paradigms, and conclude with a call to action on research steps to be taken to make verifiable purpose limitation a standard for industrial AI, ensuring it is as private as it is productive.

2. A Framework for Verifiable Purpose Limitation

We now formalize the sensing pipeline outlined in the introduction. A camera captures raw visual data X , which a privacy filter f_θ transforms into a task-specific representation $\tilde{X}$. This filtered output alone is transmitted for downstream processing (e.g., on a cloud service or third-party API) to perform the declared task T , yielding predictions Y_T .

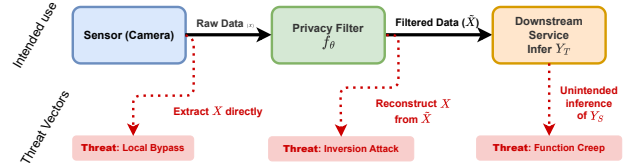


Figure 1. Threat model illustrating the sensing pipeline and potential adversaries. The camera captures raw data X , which is processed by the privacy filter f_θ to produce $\tilde{X}$. The downstream party can attempt function creep, while an adversary with access to f_θ can attempt inversion attacks. Local bypass attempts target the edge device directly.

2.1. Adversary Model

Figure 1 illustrates the sensing pipeline and potential adversaries. We consider three hypothetical adversaries with varying capabilities:

1. **Downstream function creep:** An honest-but-curious or malicious downstream party receives filtered outputs $\tilde{X}$ and attempts non-consented inference by applying unauthorized models or statistical inference to repurpose data for unintended tasks S .
2. **Inversion attack:** An adversary with white-box access to the deployed filter f_θ attempts to reconstruct or invert raw inputs X from the filtered representation $\tilde{X}$.
3. **Local bypass:** An attacker on the edge device attempts to circumvent the filter f_θ to access raw sensor data X directly.

The first two adversaries exploit residual information retained in $\tilde{X}$ and are thus addressed through strict data minimization at the output of f_θ . The third targets the filter deployment itself and is countered through hardware-enforced edge execution, as developed below.

2.2. Formalizing Minimization via Information Constraints

Both downstream function creep and inversion attacks are fundamentally enabled by the retention of extraneous information in the filtered output. To neutralize these threats, we advocate for strict data minimization at the point of capture. While several empirical learning strategies attempt to achieve this, such as adversarial training (Huang et al., 2017; De Coninck et al., 2024; 2026), disentangled representation learning (Wang et al., 2024) or optical feature separation (Boloor et al., 2025), they all share a common implicit goal: maximizing utility for a specific task while minimizing the representation’s overall information footprint. To translate this abstract goal into a concrete, enforceable system constraint, we propose formalizing data minimization

through the Information Bottleneck (IB) principle (Tishby et al., 2000).

As established, our goal is to learn a privacy filter f_θ , parameterized by a neural network, that produces a transformed representation $\tilde{X} = f_\theta(X)$. Ideally, $\tilde{X}$ should preserve only the information necessary for a declared task Y_T (e.g., pose estimation) while strictly suppressing all other information (e.g., identity, gender, background).

Regardless of the specific training algorithm used, the objective of a privacy-preserving system is to optimize the trade-off expressed by the Information Bottleneck functional:

$$\min_{\theta} \mathcal{L}(\theta) = \min[H(Y_T | \tilde{X}) + \beta I(\tilde{X}; X)], \quad (1)$$

where $H(\cdot | \cdot)$ denotes conditional entropy and $I(\cdot; \cdot)$ mutual information. In this context, Equation (1) serves as the formal definition of compliance: the first term enforces operational utility, while the second mathematically penalizes information retention beyond what is strictly required.

We view this formulation as the operational definition of purpose limitation. Systems employing adversarial learning or disentanglement are effectively seeking to approximate this bound. By adopting this information-theoretic definition, we can derive requirements that are agnostic to the specific model architecture but rigorous in their privacy guarantees. In practice, $I(\tilde{X}; X)$ is intractable for high-dimensional inputs; implementations use variational bounds (Tishby & Zaslavsky, 2015) or proxy objectives such as adversarial suppression losses. However, the principle of minimizing this mutual information remains a clear and enforceable standard for purpose limitation, providing a concrete target.

This formulation directly dictates two key system-level requirements:

- **Requirement 1: Utility-Only Supervision.** Existing privacy-preserving vision methods frequently rely on labeled sensitive attributes Y_S (e.g., identity or demographic labels) during training (Zhao et al., 2025). In industrial contexts, collecting such labels may be legally restricted or operationally infeasible (Andrews et al., 2023; Shahriar et al., 2023). The operational definition in Equation (1) avoids this dependency: the filter is trained solely with respect to the declared task Y_T , suppressing information unrelated to Y_T without requiring explicit annotation of sensitive attributes.
- **Requirement 2: Holistic Full-Frame Minimization.** Many deployed privacy mechanisms target specific visual cues (e.g., faces), implicitly assuming a narrow threat model. Industrial environments expose broader risks, including soft biometrics and proprietary spatial layouts. Minimizing the mutual information $I(\tilde{X}; X)$

enforces a global constraint on information flow, ensuring no visual feature, local or distributed, is retained unless it contributes to task performance.

2.3. Architectural Modularity and Edge Enforcement

Beyond theoretical guarantees, strict minimization must integrate with existing industrial machine learning pipelines. We therefore advocate for an architecture in which the privacy filter is implemented as a *modular frontend* that precedes the task model:

$$Y_T = M_{\text{frozen}}(f_\theta(X)). \quad (2)$$

Here, M_{frozen} denotes a pre-trained and validated task model, which remains unchanged.

This design choice enables two additional requirements.

- **Requirement 3: Task Model Preservation.** Requiring retraining or modification of deployed models represents a significant barrier to adoption (Agrawal et al., 2021). Keeping the output of f_θ in an image-like format enables plug-and-play reuse of existing image-processing pipelines and pretrained models. This modularity also supports independent auditing, updating, or replacement of the privacy mechanism, and eases integration with third-party services (e.g., MLaaS/RaaS) without vendor lock-in.
- **Requirement 4: Edge-Native Enforcement.** To provide a technical enforcement mechanism, the privacy filter must be deployable at the edge, ideally within a TEE on camera. This directly counters the local bypass adversary: raw sensor data are irreversibly transformed at the point of capture and never leave the device in identifiable form, eliminating the attack surface before any circumvention attempt can succeed. Importantly, while downstream models may leverage large-scale pretraining (Tramèr et al., 2024), the privacy filter itself must remain lightweight and task-specific such that it can be deployed in resource-constrained TEEs. In cloud-based inference and third-party deployments, this architecture reduces what leaves the device regardless of provider, narrowing the trust and attack surface.

2.4. Envisioned Deployment Scenario

Returning to the scenario from the introduction: cameras are installed for ergonomic assessment via human pose estimation. The organization first specifies the task T and obtains consent from relevant stakeholders for this purpose. Cameras are equipped with an on-device data minimization filter f_θ , trained to retain only pose-relevant information, with the aid of the task model M_{frozen} , and deployed within a

TEE. A third-party auditor verifies both the implementation and the declared task alignment.

Only the filtered output $\tilde{X}$ is transmitted to an on-premise system or external service, where pose estimation is performed using a frozen, pre-validated model M_{frozen} . If additional functionality is later proposed (e.g., quality control), a new task declaration, consent process, and corresponding minimization filter are required prior to deployment. Figure 2 provides a conceptual overview of this architecture and compares it to traditional redactive obfuscation.

Learning the privacy filter f_θ requires access to representative raw data and task labels during the development phase, which we treat as a distinct and complementary challenge from the continuous, unbounded surveillance risk of deployed systems. We view training-time privacy as a complementary requirement that is well-addressed by existing literature, including Federated Learning (Zhang et al., 2021) and differential privacy (Dwork, 2008). While these methods minimize risk during development, our position addresses the operational necessity: ensuring that the final deployed artifact transmits only information necessary for the declared task, with unnecessary or sensitive information systematically suppressed at the point of capture.

3. Limitations of Existing Inference-Time Privacy Mechanisms

Privacy-preserving computer vision research is typically categorized into three distinct stages (Miresghallah et al., 2020): (i) data publishing, (ii) model training, and (iii) inference and deployment. While significant progress has been made in the first two stages, through techniques like differential privacy, federated learning, and dataset distillation, these methods primarily protect the training data or the model parameters. They do not, however, constrain the sensing capabilities of a deployed system. A camera with a model trained via federated learning still captures high-resolution, identifiable footage at runtime, leaving the surveillance risk at the edge unmitigated.

Therefore, this section focuses exclusively on inference-time privacy: techniques that operate at or immediately after the time of capture. We argue that the prevailing digital approaches fall short of responsible industrial AI because they focus on concealment (hiding sensitive pixels) or confidentiality (encrypting the stream) rather than minimization. In the following sections we go through the main categories of inference-time privacy techniques, highlighting their limitations in terms of data minimization and function creep protection. We conclude with a summary of the key gaps that motivate our proposed approach.

3.1. Redactive Obfuscation: Blurring and Inpainting

Redactive obfuscation remains the most common visual privacy strategy, which typically identifies sensitive regions (e.g., faces) and applies blurring, pixelation, or inpainting (Zhao et al., 2025). These methods are intuitive and easy to explain to non-expert stakeholders, including employees, which contributes to their widespread adoption. However, they are fundamentally reactive and limited by a detection-first bottleneck: only what is successfully detected can be protected (Hukkelås et al., 2023; Cangialosi et al., 2022). In industrial environments, occlusions, motion blur, and poor lighting frequently cause false negatives, where even a single missed detection can result in an irreversible privacy leak. Furthermore, the approach requires operators to pre-define an exhaustive list of sensitive attributes; any category omitted from this specification (body proportions, tools, background layout) is silently retained in full.

Additionally, these techniques are typically task-agnostic, degrading both sensitive attributes and information required for downstream analysis (Lee & You, 2024). For example, pixelation may conceal identity while simultaneously obscuring joint-angle cues needed for ergonomic assessment, creating an unnecessary trade-off between privacy and utility that conflicts with the data minimization principle.

Even when detection succeeds, simple visual distortions provide limited protection against modern learning-based adversaries, which can reconstruct identities using learned priors (Zhai et al., 2026). Moreover, abstraction fails to remove soft biometrics such as gait or body proportions, enabling re-identification even without facial information (Oh et al., 2016; Weiß et al., 2026). By treating privacy as post hoc redaction rather than a system-level constraint, these approaches remain vulnerable to both reconstruction attacks and unintended leakage of biometric or proprietary information.

3.2. Differential Privacy

Differential Privacy (DP), originally developed for database queries (Dwork, 2008), has been extended to Local Differential Privacy (LDP) and metric privacy to support high-dimensional visual data. By providing indistinguishability guarantees proportional to input distance, metric privacy enables privacy-preserving inference in continuous image and video spaces. Prior work focuses largely on facial privacy, including GAN-based private image generation (Croft et al., 2022), semantic identity masking (Chen et al., 2021), and video extensions incorporating temporal notions such as event duration (Cangialosi et al., 2022) or edge-based split-model architectures for live streaming (Nagasubramaniam et al., 2024).

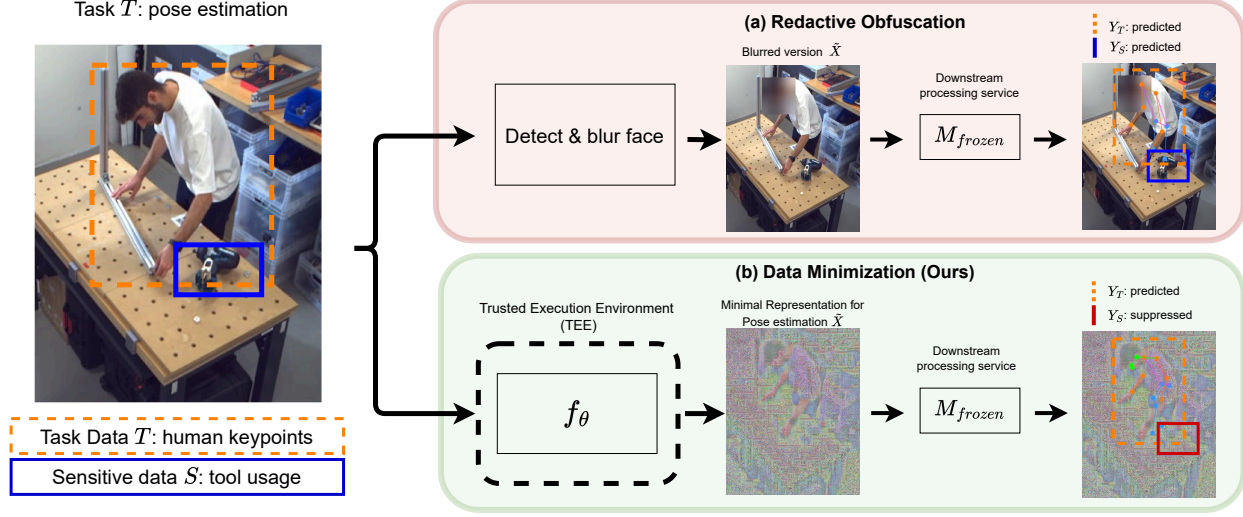


Figure 2. Comparison of redactive obfuscation (a) and task-bound data minimization (b) for industrial pose estimation (Y_T), with tool usage as the sensitive attribute (Y_S). In (a), the operator must explicitly define sensitive regions (here: face); Y_T is preserved but Y_S remains detectable in $\tilde{X}$, as any unspecified attribute is silently retained. In (b), f_θ runs inside a TEE and requires no sensitive labels: Y_T is preserved while Y_S is architecturally bounded by design. Images and filtered representations are adapted from De Coninck et al. (2026) and are illustrative of the qualitative effect.

However, these approaches emphasize feature indistinguishability rather than data minimization. DP protects sensitive attributes within collected data but does not prevent over-collection at capture time, a particularly challenging issue in computer vision, where multiple semantic concepts are tightly entangled within a single frame. In industrial settings, data minimization and DP should be complementary: task-relevant information should be isolated at capture, while DP provides formal guarantees over the remaining attributes. Existing methods rarely offer such integration, often operating on full visual inputs before any privacy mechanism is applied, or applying privacy at later stages such as intermediate features, latent representations, or model updates.

3.3. Cryptographic Methods and Trusted Execution Environments

Cryptographic approaches such as Homomorphic Encryption (HE) and Secure Multi-Party Computation (SMPC) provide confidentiality-in-use by enabling inference directly on encrypted data. Although recent systems extend these techniques to complex models, including vision transformers (Pang et al., 2024) and diffusion models (Zhao et al., 2024), they remain heavily constrained by non-linear operations (Pindado et al., 2026). Functions central to attention mechanisms, such as Softmax and GELU, incur orders-of-magnitude higher computational cost than standard convolution, severely limiting practicality.

Trusted Execution Environments (TEEs) face complemen-

tary limitations at the edge: CPU-based enclaves (e.g., Intel SGX, ARM TrustZone) impose strict protected-memory caps that hinder large-model execution and require costly partitioning (Mo et al., 2020). While confidential computing is emerging on GPUs, support is largely confined to data-center-class hardware (e.g., NVIDIA H100) (NVIDIA Corporation, 2023), creating a deployment gap in which secure inference is feasible only in the cloud, forcing sensitive data off-device.

More fundamentally, cryptographic and hardware-based protections do not satisfy data minimization. These mechanisms regulate who may access or process data, but remain agnostic to what information is collected and represented. While they safely secure data against unauthorized third parties, HE, SMPC, and TEEs offer weak technical barriers against first-party function creep: the entity holding the decryption keys or controlling the TEE can easily authorize secondary, non-consented inferences later. As a result, they secure computation without enforcing purpose limitation, leaving prevention of function creep dependent on external policy and trust in the system owner rather than end-to-end system guarantees.

3.4. Optical Privacy Filters

Optical privacy represents a paradigm shift in computer vision, moving from post-processing obfuscation to physical, hardware-level protection at the point of capture. By integrating end-to-end learning with optical design, researchers have developed specialized lenses that facilitate task-specific

performance, such as human pose estimation (Hinojosa et al., 2021), while intentionally degrading identity-sensitive features. This approach extends to 3D sensing through the use of learned phase masks for passive depth estimation (Tasneem et al., 2022), ensuring that depth maps are generated without recording high-fidelity imagery. Furthermore, frameworks like PrivHAR demonstrate that complex tasks such as human action recognition (Hinojosa et al., 2022) can be performed effectively through privacy-preserving lenses that filter out sensitive details. Finally, PrivateEye (Boloor et al., 2025) takes this further by employing a diffractive optical neural network to spatially route task-relevant and privacy-sensitive features to distinct sensor regions, reading out only the utility pixels so that private information is never digitized at the edge device. Optical privacy thus offers a compelling solution to the data minimization challenge; it is relatively up and coming and currently still limited in its flexibility compared to digital methods (Boloor et al., 2025).

3.5. Summary

We summarize the key differences between these inference-time privacy approaches in Table 1. Across categories, a common shortcoming emerges: most methods protect *access* to data or redact it post-capture, whereas optical privacy methods are a promising exception that begins to constrain *what* is captured. This remaining gap motivates inference-time mechanisms that enforce strict data minimization while remaining adaptable to real-world deployment constraints.

4. Alternative Views

Having a system based on purpose limitation does not necessarily make it private. What about scenarios where the intended task is inherently correlated with sensitive information?

This concern is well-founded. In some cases, the declared task T is inherently correlated with personal information. For example, a system designed to analyze an individual’s gait for safety purposes may also enable identity inference, as gait patterns are person-specific. However, we argue that this makes architectural limits more important, not less. Without the technical constraints we propose, workers must implicitly trust that a system captures no more than declared. By enforcing that the system retains only the features sufficient for the task, a verifiable boundary is established. While residual correlations may permit some sensitive inferences, the system is structurally prevented from collecting extraneous information. This provides workers and organizations with a transparent basis for consent: agreement is scoped to a specific, bounded process rather than open-ended surveillance. We discuss how to quantify these residual correlations in Section 5.4. Our claim is therefore not that purpose

limitation alone ensures privacy, but that it is a necessary architectural constraint for bounding and making explicit the privacy risks that remain.

In industrial settings, performance comes first. Privacy filters reduce data quality and accuracy, but automation and monitoring require precise, high quality data, making any loss in reliability unacceptable.

We acknowledge the engineering validity of this perspective; retaining maximum information theoretically ensures models remain robust and adaptable to undefined future tasks. However, in workplace settings, this surplus of information creates a critical tension between technical potential and social acceptance. Ignoring this constraint leads to a deployment paradox: a system with high theoretical utility yields effectively zero real-world value if it is rejected by workers or blocked by compliance teams due to surveillance concerns (Sebastian et al., 2025). Thus, we view strict purpose limitation not as a hindrance to innovation, but as a necessary enabler. It resolves the trust bottleneck, allowing safety technologies to be deployed in sensitive environments where they would otherwise be socially or legally untenable (Leesakul et al., 2022).

You are putting technology first, not the user. The most privacy preserving approach is to not capture any data at all, if it could infringe on privacy.

Eliminating data collection does eliminate many categories of privacy risk. However, outright rejection of computer vision systems on privacy grounds also forgoes systems that protect workers from physical harm, such as those that prevent accidents or monitor ergonomic strain to reduce long-term injury. We contend that a genuinely worker-centric approach need not require a choice between physical safety and informational privacy. Rather than precluding the technology, the appropriate response is to design the system such that over-collection is structurally impossible. With verifiable data minimization, the worker’s right to a safe and private workplace is the starting constraint, privacy stops being an obstacle to deployment and becomes the reason the system can be deployed at all. Our position is scoped to settings in which vision-based systems are already under consideration or deployment, and we do not advocate for their universal application.

Performing all computations on-device in a secure environment is a safer and reliable alternative

Edge processing is undeniably effective for mitigating data-in-transit risks and handling latency-sensitive tasks. However, the rapid industry consolidation around large foundation models presents a new challenge: these architectures

Table 1. Comparison of inference-time privacy approaches.

Feature	Detection-Based Heuristics	Encryption & TEEs	Differential Privacy	Optical Privacy Filters	Strict Data Minimization (Proposed)
Mechanism	Selective region obfuscation	Encrypted or enclave-based inference	Noise injection (pixel or feature space)	Physical lens or sensor modification	Information Bottleneck (IB) extraction
Enforcement Mode	Post-capture obfuscation	Confidentiality-in-use	Statistical indistinguishability	At-sensor physical filtering	Edge-native enforcement (TEE)
Function Creep Protection	Weak: vulnerable to reconstruction	Weak: relies on key-holder policies	Partial: obscures individuals, not concepts	Strong: no high-fidelity capture	Strong: limited use for secondary tasks
Data Minimization	Weak: limited to detected regions	None: underlying semantics preserved	Partial: noise does not remove all attributes	Strong: physical omission at capture	Strong: enforced by design
Primary Limitation	Detection reliability	Latency or memory constraints	Utility degradation	Hardware inflexibility	Leakage–utility auditing

frequently exceed the computational and memory bounds of consumer hardware (Rodriguez-Conde et al., 2021), effectively putting state-of-the-art performance out of reach for purely local devices. Furthermore, while keeping raw data on-device is often viewed as the gold standard for privacy, it is not impervious to attack. Even distinct in-memory processes remain susceptible to exfiltration via insecure APIs or inference-time side channels (Baldassini et al., 2025), requiring users to place implicit trust in the device’s firmware integrity. Strict data minimization offers a robust alternative to this binary choice. By anchoring the minimization logic within a TEE, this approach mathematically restricts information at the point of capture, combining the verifiable safety of local filtering with the performance of cloud-based models.

Instead of using cameras at all, it’s much better to look at alternative sensors that are inherently privacy-sensitive, such as infrared, lidar, radar etc.

A commonly proposed approach is to reduce or avoid the use of cameras by relying instead on sensing modalities often perceived as more privacy-sensitive, such as infrared (Dong et al., 2024), LiDAR (Günter et al., 2020), or radar (Mineo et al., 2025). This view has merit: these modalities capture less visually recognizable information than RGB cameras, and the results are harder for humans to interpret intuitively, which tends to read as more private. However, this intuition can be misleading, as the informational content of such signals is not readily accessible to human judgment, making it difficult to assess what sensitive attributes may still be inferred, particularly when combined with modern machine learning methods. Recent work shows that these modalities can still enable re-identification or the inference of sensitive information (Baldassini et al., 2025; Lintvedt, 2023; Liu

et al., 2025).

Beyond privacy considerations, alternative modalities involve practical trade-offs, including limited spatial resolution for infrared imaging (Hou et al., 2022), higher cost, and sensitivity to environmental conditions (Sezgin et al., 2023). Moreover, many industrial settings already rely on extensive camera infrastructure (Dotoli et al., 2019), and the broader machine learning ecosystem remains largely centered around RGB imagery, making large-scale replacement costly and disruptive. Taken together, these factors suggest that infrared, LiDAR, and radar sensors are best viewed as alternative sensors to which strict data minimization principles can also be applied, rather than as outright replacements for camera-based systems.

5. A Call To Action

5.1. Context-Aware Privacy Standards

Privacy is not an intrinsic property of data, but a relationship governed by the environment and available information flow (Nissenbaum, 2004; Wishart et al., 2007). While strict data minimization is the most robust technical baseline, the safety of the remaining information fluctuates based on external context. A minimized feature representation might be anonymous in isolation, but can become personally identifiable when cross-referenced with auxiliary modalities, such as textual metadata or worker shift logs (Ding et al., 2025). Consequently, we argue that privacy frameworks cannot be static; they must quantify how the availability of contextual information alters the inferential capacity of the data. Future research should focus not on complex dynamic enforcement, but on context-aware risk metrics that explicitly model the adversarial utility of side-information. Mapping these de-

dependencies rigorously would let us give stakeholders an honest picture of residual risk, so consent covers what the system actually does, not just what the camera is pointed at.

5.2. Hardware-Rooted Privacy Integrity

To move from policy-based promises to technical certainty, privacy mechanisms must be anchored in hardware-rooted trusted execution environments, ensuring the transformation from raw pixels to task-specific features is immutable and isolated from OS-level tampering. However, a critical gap exists between the secure-but-constrained CPU enclaves available today and the high-performance confidential GPUs restricted to data centers (NVIDIA Corporation, 2023). While architectures like DarkNetZ (Mo et al., 2020) demonstrate the potential of split-computing, they are severely bottlenecked by the limited memory and compute throughput of commodity standards like Intel SGX (Costan & Devadas, 2016) and ARM TrustZone (Pinto & Santos, 2019), and current TEEs remain vulnerable to side-channel attacks such as TEE.fail (Chuang et al., 2026). Consequently, we propose a two-pronged research agenda. First, the community must focus on algorithmic optimization for existing TEEs, designing lightweight, quantized filter models capable of running within the strict resource bounds of current CPU enclaves. Simultaneously, there is an urgent need for the development of confidential edge accelerators, extending TEE boundaries to embedded GPU platforms (e.g., NVIDIA Jetson class) to enable high-bandwidth, tamper-proof video analytics at the physical point of capture.

5.3. Bridging the Socio-Technical Gap

Technical robustness alone is insufficient; if stakeholders, from resource-constrained SMEs to individual workers, do not understand or trust a system, its deployment faces significant barriers (Distler et al., 2020). We argue that the implementation of strict data minimization requires effective communication regarding the nature of the data being captured. It is not enough to minimize data mathematically; stakeholders must be able to comprehend what information remains and why it adheres to the principle of purpose limitation. Consequently, the research community should prioritize the development of intelligible privacy interfaces that translate abstract feature representations into accessible concepts, allowing non-experts to verify the boundaries of data capture for themselves (Esmailzadeh et al., 2019). By ensuring that the scope of minimization is transparent rather than opaque, we facilitate the trust required for sustainable adoption (Mannhardt et al., 2019).

5.4. Quantifying Semantic Correlation

We call on the research community to move beyond simple “hiding” and begin to rigorously quantify the intrinsic cor-

relations between task signals and sensitive attributes. To fulfill the promise of transparency discussed in Section 4, stakeholders require a “Privacy Impact Audit” that proves task-specific data does not mathematically imply the presence of protected information beyond an acceptable threshold.

A representative tension is monitoring worker fatigue via facial analysis while masking identity: features that capture fatigue (e.g., eyelid droop) can often be repurposed for re-identification or demographic inference (Dantcheva et al., 2015). The rise of large multimodal models further expands the surface for such unintended inference (Tömekçe et al., 2024). While emerging datasets now contain extensive privacy-relevant annotations (Xu et al., 2024; Orekondy et al., 2017), these resources must be actively leveraged to systematically map the hidden dependencies between visual concepts. Crucially, this quantification is the prerequisite for the transparency discussed above; one cannot truthfully communicate the risks of data sharing without first auditing exactly which protected attributes are correlated with the consented task (Zhang et al., 2023).

5.5. Operationalizing Purpose Limitation

To make purpose limitation deployable in practice, the community must address three operational challenges. First, task-specific privacy requires efficient execution of multiple filters under the strict memory and latency constraints of edge TEEs, where prior work already shows that secure inference is feasible but resource-bound (Mo et al., 2020; Steiakakis & Vasiliadis, 2026). This calls for lightweight, quantized, or partially shared multi-task designs that preserve task-specific minimization without making real-time deployment impractical (Alvar & Bajić, 2021). Second, privacy audits must become standardized and continuous: filtered outputs should be routinely stress-tested with attribute-inference and reconstruction probes, with explicit acceptance thresholds and re-audit triggers after model or sensor updates (Zhang et al., 2023; Baldassini et al., 2025). Third, these audits must remain valid under deployment drift, since changes in lighting, viewpoint, or context may alter both utility and privacy behavior in ways that are not visible from task accuracy alone. Without progress on these three fronts, purpose limitation stays a well-defined goal with no practical path to enforcement.

6. Conclusion

Industrial computer vision poses a genuine privacy problem that blurring faces and writing data-handling policies cannot solve. As long as raw visual data flows beyond the camera, the system retains the capacity for inferences that workers never consented to — regardless of what the operator intends.

The alternative we have argued for is architectural: restrict what the camera produces, not just what downstream parties are permitted to do with it. Optical privacy filters establish that constraining sensing at the point of capture is physically achievable; the Information Bottleneck formalizes that principle as a learnable, task-specific objective; and Trusted Execution Environments provide the hardware anchor that makes the guarantee verifiable rather than declaratory. The result is a pipeline where function creep is structurally bounded, not merely discouraged.

What the framework needs to become deployable is still being built: lightweight filters within TEE memory budgets, audit protocols that survive model updates and sensor drift, and interfaces through which workers can meaningfully verify what the system retains, including where task signals and sensitive attributes remain partially entangled. We have tried to be concrete about what each of these requires. Data minimization cannot close every gap, but it can make the remaining ones visible.

Acknowledgements

Sander De Coninck receives funding from the Special Research Fund of Ghent University under grant no. BOF22/DOC/093. This research is done in the framework of the Flanders AI Research Program (<https://www.flandersairesearch.be/en>).

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